Day 2

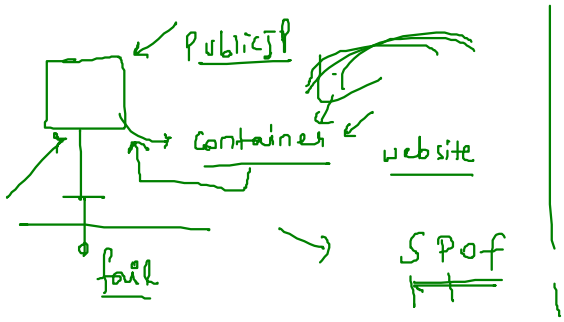
Recap :-

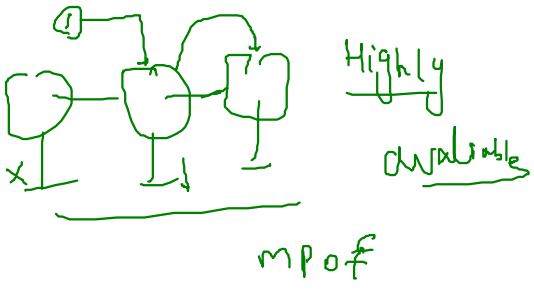
1. Basics Linux
2. Why Containerization -- 2013
3. Architecture of Docker
4. Docker images , docker containers
5. DOwnload the image
6. Create the containers
7. confguring apacge server in containers
8. Port Mapping

Todays Agenda :-

1. Docker Disadvantages
2. Why Kubernetes came?
3. Architecture of Kubernetes
4. Components of Kubernetes
5. INstall the kubernetes
6. Access Kubernetes
7. Resources of Kubernetes
    a. Namespace
    b. Pods
    c. Deployments

8. Why Openshift Came
9. Openshift Architecture
10. Components
11. How to access Openshift Cluster
12. Resources of Openshift
    a. Namespace
    b. Pods
    c. Deployments
    d. Services
    e. Routes

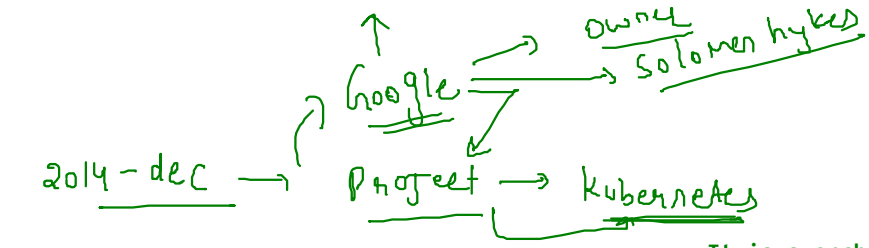




1

Docker is a tool to only create containers

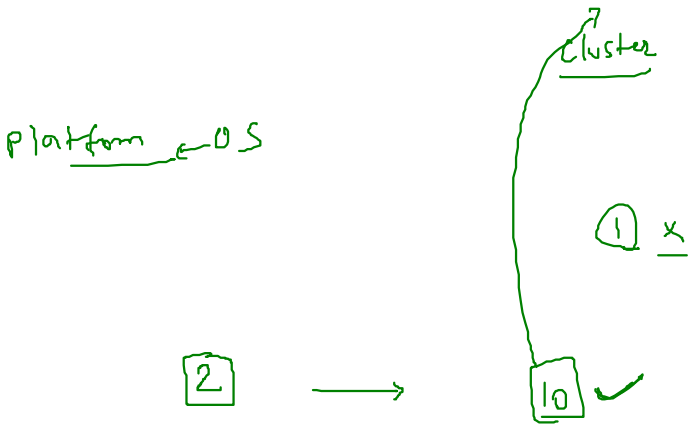
2

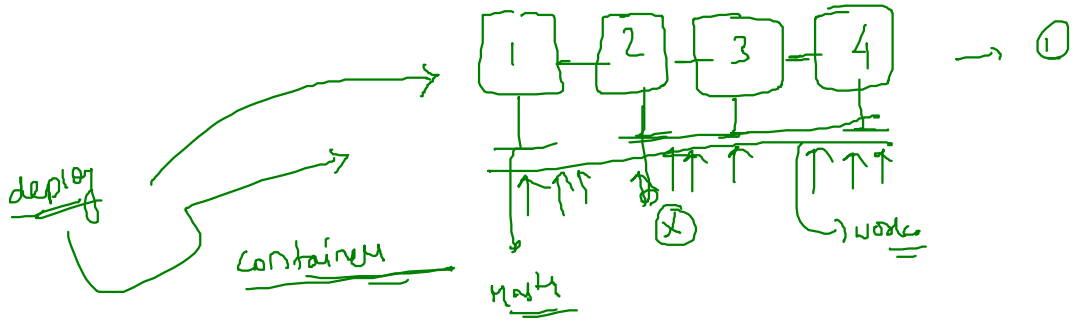


It is a orchestration/Manager/Management tool for contai

Docker is a tool to createcotnainers
Kubernetes is tool to manage containers

Indirectly kubernets will help to create containers





Master / Worker

1 M → Req → worker

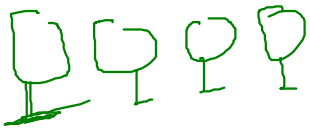
↓

SPof

multi master Kubernetes

↓

LB



Components of kubernetes

use → container → Pod → testpod

1 kube Api server → Apiresources

2 ~~Kube~~ etcd - server key=value

3 kube Scheduler

Pod → container

Pod → resource NAME

CAPTAIN

RAM CPU Stor

install captain

kubelet

Pod → downloads image → Contain

Single Node cluster

Inits Pods

kubeproxy

CoreDNS

init

mask

API
Server
etcd
Proxy

4

1. kubelet
2. ~~kubeadm~~ kubeadm
3. kubectl
4. crio / docker

Prereq:

1. swap memory ✓
2. br_netfilter ✓
3. kubelet, kubectl, kubeadm, crio

→ Script

kube init ✓

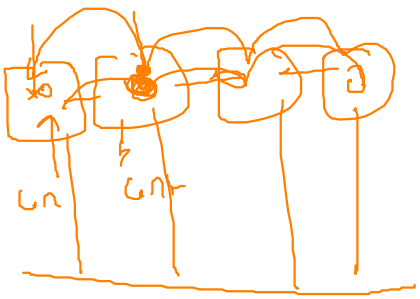
Login → US
Password ✗

certificate Based login

admin.conf → Credentials → login

~~kubed~~ kubectl → home/.kube/config

Calico
weave-net
flannel



by n/w

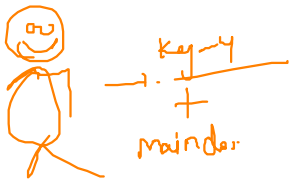
1. kubectl ✓

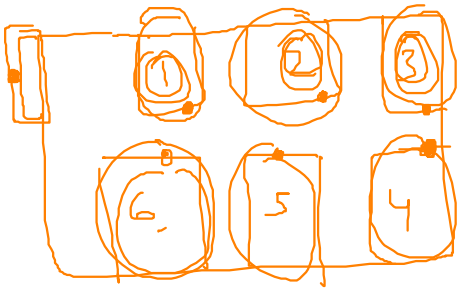
2. root → .kube

3. Config → Authentication fail

4. kubectl get node

Namspace





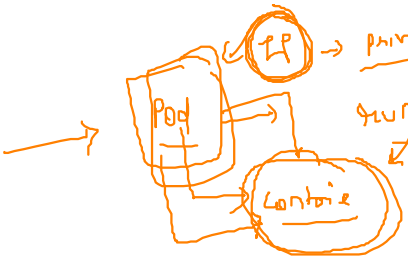
NS ✓

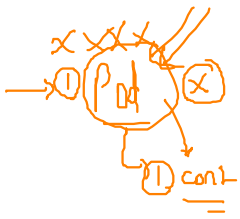
create NS

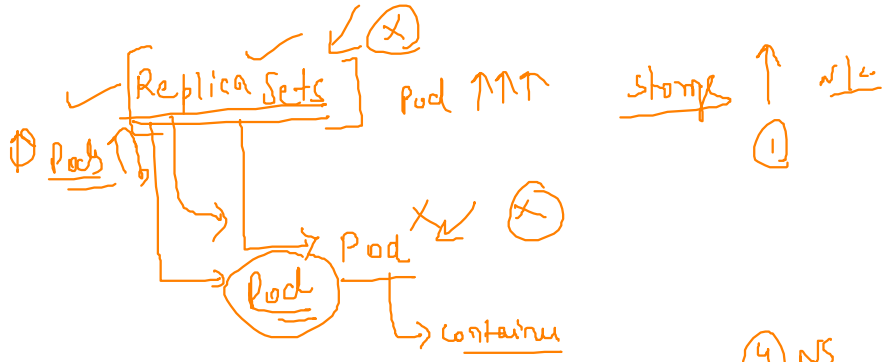
default

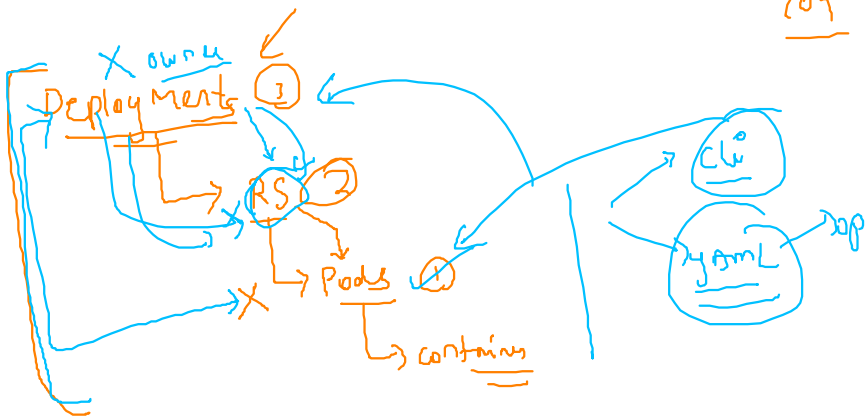
IP → private → 192.168.0.0/16

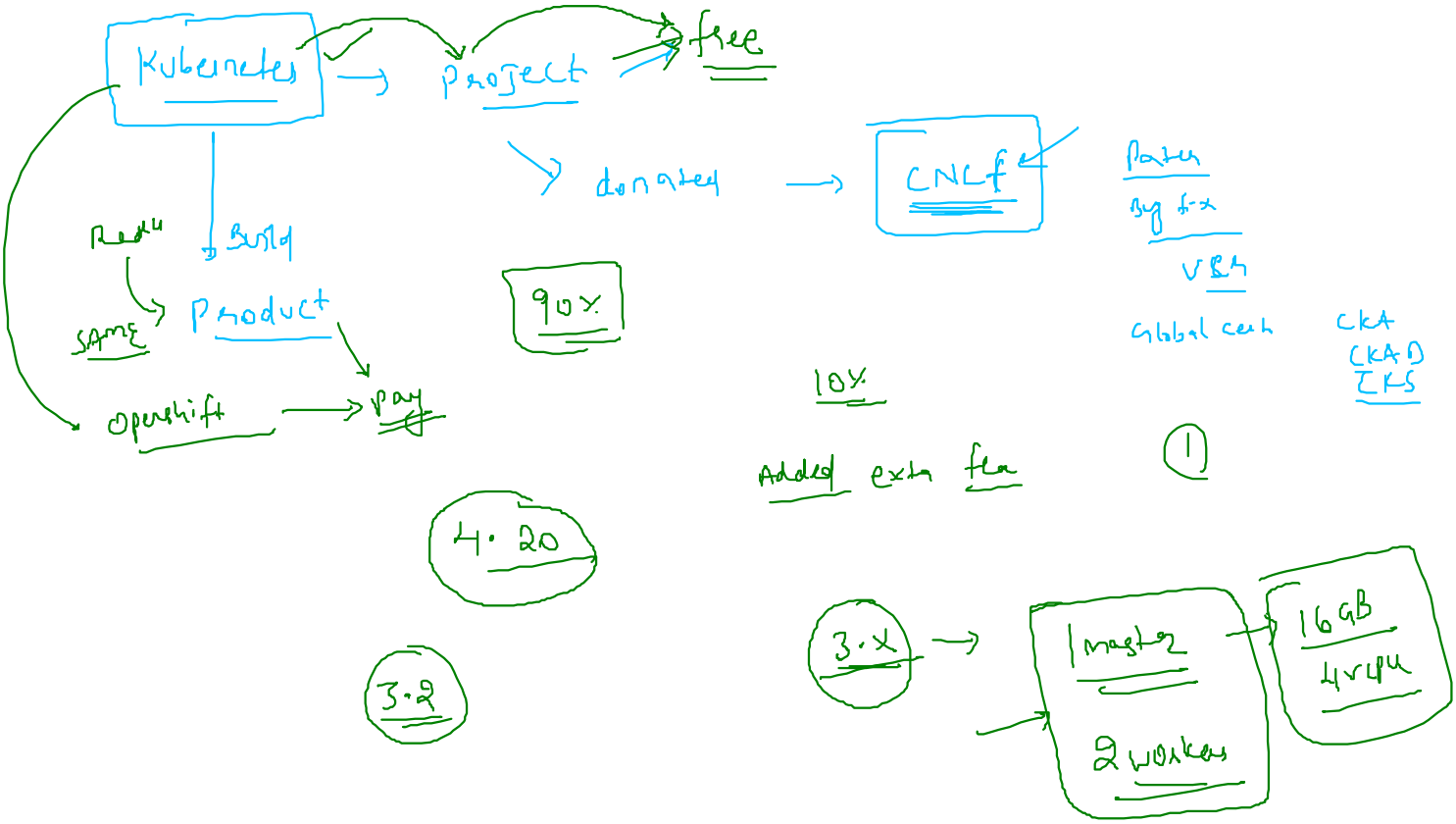
run web site

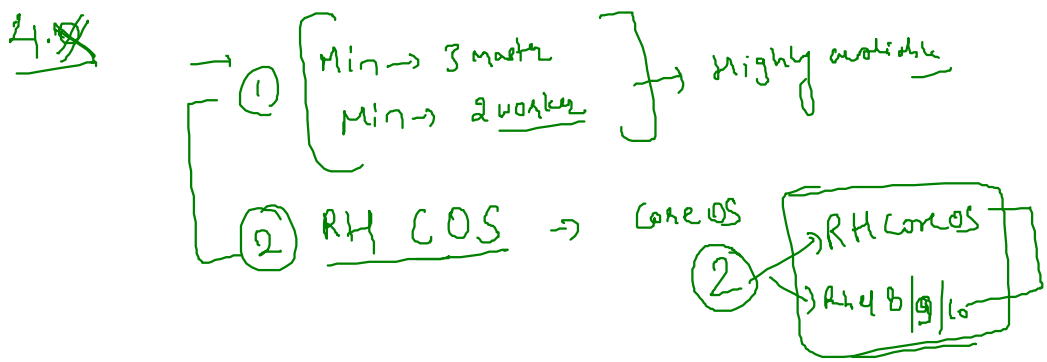





Replica Sets
Pod
Pods
Pod
Pod
containers
stomps
n/c


4 NS
Resources
Tot
Deployments
owner
RS
Pods
contains
CLI
yaml
op


Kubernetes
project
free
donated
CNCf
Paten
Big fix
VBA
Global cash
CKA
CKAD
CKS
Redhat
Build
Product
SAME
90%
10%
Added extra fea
1
openshift
pay
4.20
3.2
3.x
1 master
2 workers
16GB
4vcpu


4.x
1
Min → 3 master
Min → 2 workers
Highly available
2
RH COS
CoreOS
2
RH CoreOS
RHEL 8/9/10

kubeApiServer

↓

etcd-server + proxy

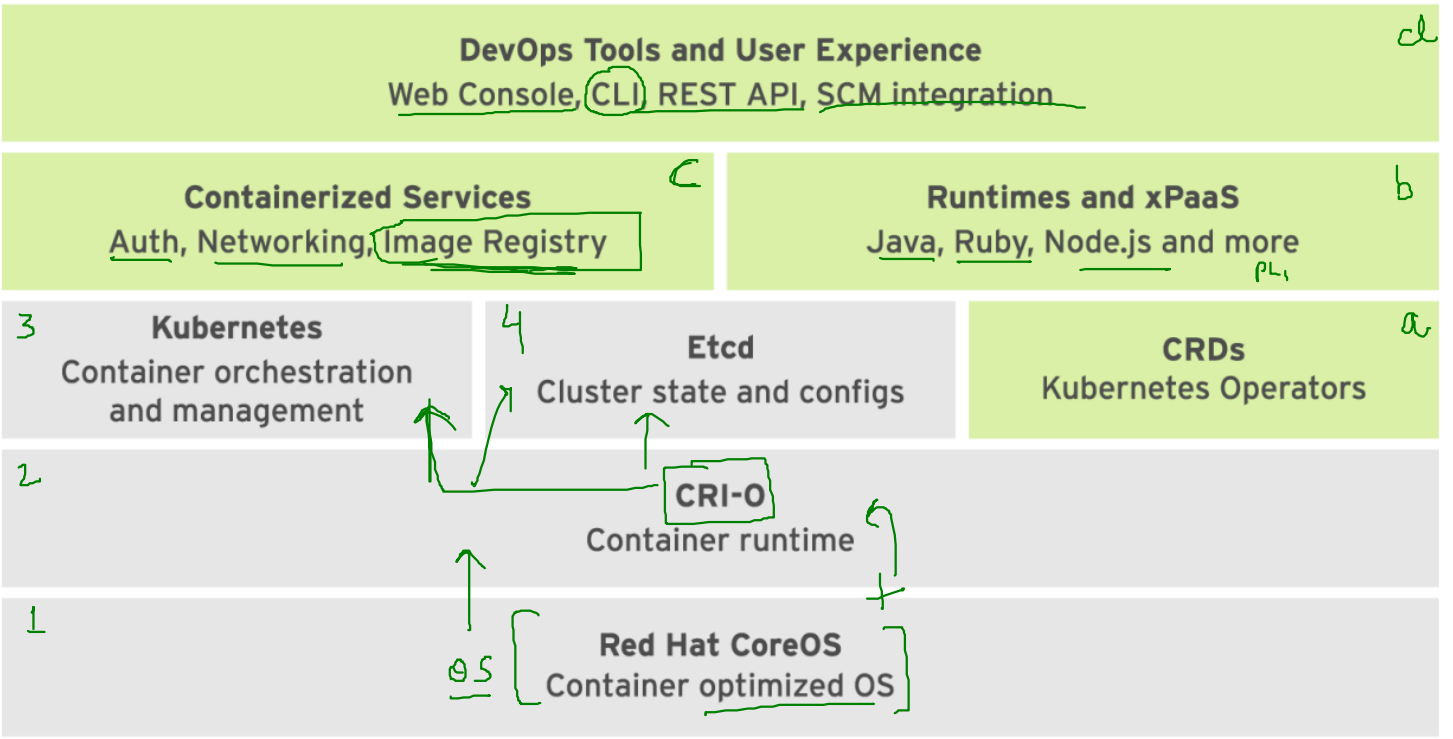
↓

scheduler

proxy. + crio + etcdser → ApiSer

+

OS



1+2+3+4 → Kubernetes

CRD →

flight

(b)

index → (1) Build custom dockerfol image → (2) Upload registry → (3) openshift

GitHub

5 mins

2 mins → 1 min.

8 mins

100%

URL → detect → image registry

to see

php

dockerfile

customimage

upload → registry

deploy

3-4 mins

Auth → (1) → Certificate

dashboard → US/Pass

CLI → US/passcode

US/Pass

DNS

XYZ.com

IP:Port

(2) N/W

App → DNS

OC

Login

Dashboard --
- a. Console URL
- b. Username
- c. Password

CLI
- a. API URL
- b. User
- c. Password

Config

NAMESPACE → Project

XXXXX

1. Login
2. overview
3. CLI → Pod / deploy → dashboard

1. Login
2. overview
3. Project / Pods / deployments ← YAML

Prose → Usern

X

Linux → root
XX

Contain → Linux Based → root

XXX root

for loop

1. from which Api
   - object → kind
   - reso

metadata
Name of pod

SPEC: Pod Contain

8 Lines → Pod

Dept = 25 lines

2-4 mins

10 mins

2 mins

edit

10 seconds

generate yaml

Services

1. ClusterIP
2. Nodeport
3. LB
4. External NAME

Deploy ← Pods → IP ← Private

Pods → IP

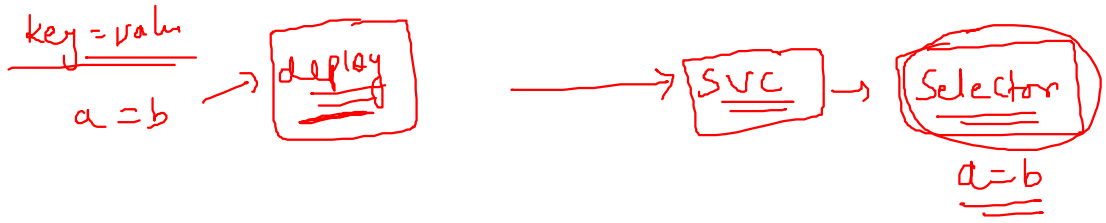
Pod → x

SVC → ClusterIP → Pod → Expose

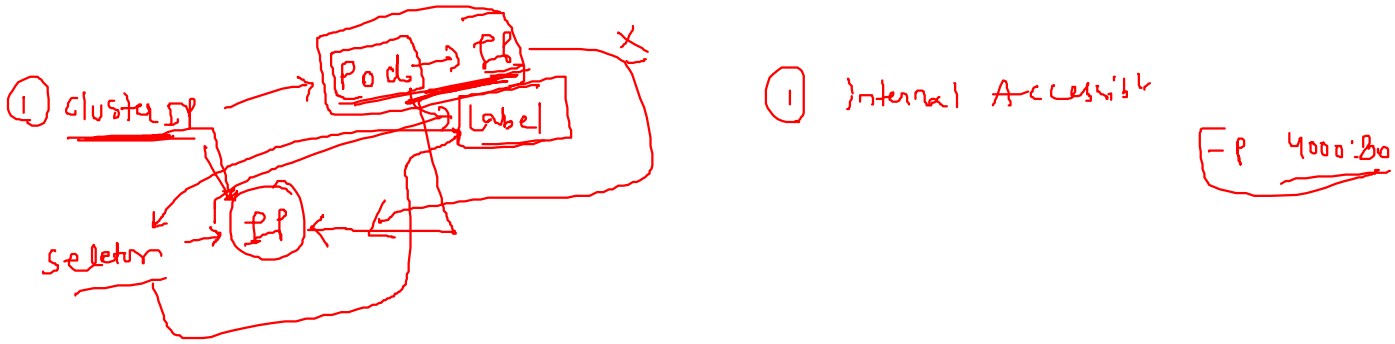
IP → Private

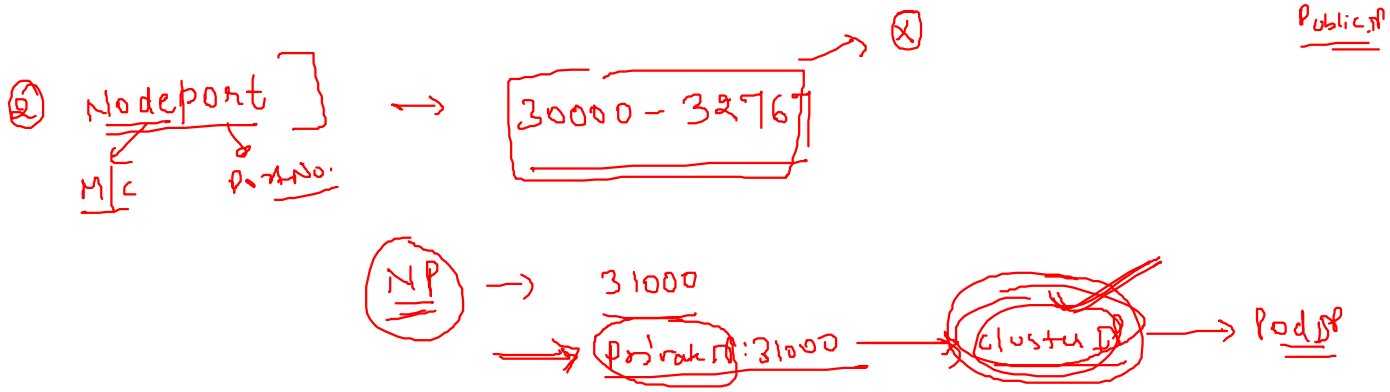
10.11.12.13 → Endpoint = Pod's IP

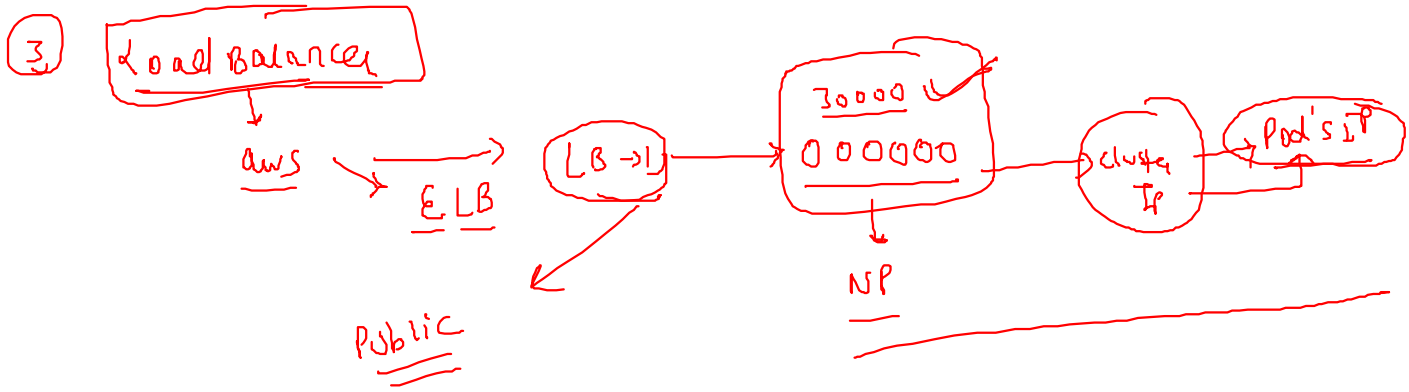
Labels & Selectors

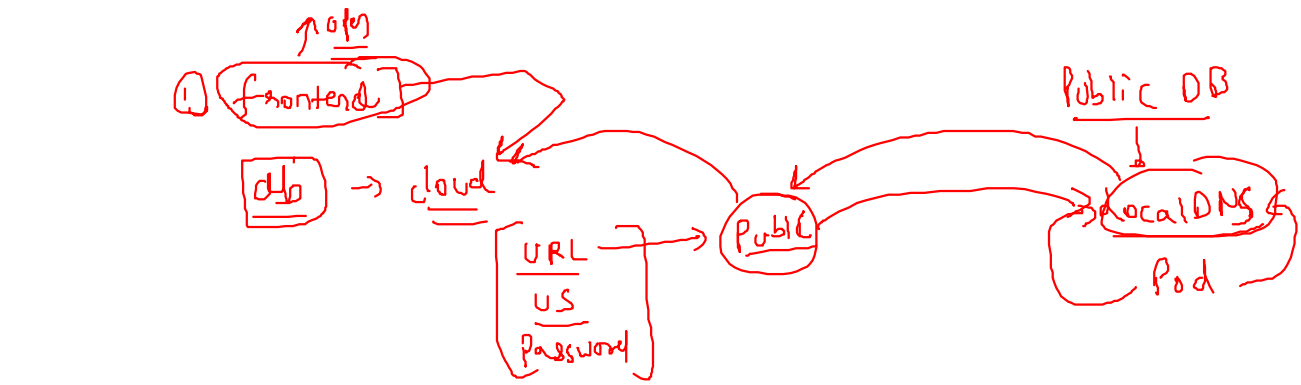


Labels of Deployment = Selectors of Service









Routes

URL →

ClusterIP

NP

Route → DNS

XYZ-com

domain

https://console-openshift-console.apps.mayank.openshiftlab.xyz

fixed

fixed

clustername

* apps.

webapp.

Pod / Depp

Svc

Routs

def

app = mayanke

key

Pod

run = Podname